Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2026

Course: B.Tech
Code: 155612

Subject: Python Programming

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper..*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which of the following statements about indentation in Python is correct?

(i) Indentation is optional and used only for readability
(ii) Indentation determines the scope/block of statements
(iii) Indentation is required only inside functions
(iv) Python uses braces {} instead of indentation

(b)
```
a = 10
b = 3
c = 2
result = a + b * c ** 2
print(result)
```
The value of result is:

(i) 676
(ii) 28
(iii) 22
(iv) 32

(c) Which statement is used to skip the remaining statements of the current iteration of a loop and proceed with the next iteration?

(i) break
(ii) continue
(iii) pass
(iv) return

(d)
```
for i in range(1, 6):
    if i == 3:
        continue
    print(i, end=" ")
```
What is the output?

(i) 1 2 3 4 5
(ii) 1 2 4 5
(iii) 1 2 3
(iv) 4 5

(e) Which statement about a Python list is correct?

(i) A list cannot contain duplicate elements
(ii) A list is immutable
(iii) A list can contain elements of different data types
(iv) A list can contain only numeric values

(f)
```
L = [10, 20, 30, 40, 50]
print(L[1:4])
```
The output is:

(i) [10, 20, 30]
(ii) [20, 30, 40]
(iii) [20, 30, 40, 50]
(iv) [10, 20, 30, 40]

(g) Which statement about a tuple is correct?

(i) Tuples are mutable
(ii) Tuples cannot contain duplicate elements
(iii) Tuples are immutable
(iv) Tuples can contain only numeric values

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(h) Which file mode opens a file for reading binary data?
(i) r
(ii) w
(iii) rb
(iv) br

(i) Which OOP concept allows a derived class to acquire properties and methods from a base class?
(i) Polymorphism
(ii) Inheritance
(iii) Compilation
(iv) Iteration

(j) Which of the following is a suitable application of NumPy?
(i) Creating HTML pages
(ii) Numerical and array-based computations
(iii) Sending emails
(iv) Managing operating-system processes only

Q.2 (a) Describe the arithmetic and relational operators in Python with suitable examples. Write a Python program to demonstrate their use. [7]

(b) Explain type conversion in Python. Differentiate between implicit and explicit type conversions with suitable code examples [7]

Q.3 (a) Write a Python program to find the largest of three numbers using appropriate decision control statements. [7]

(b) Explain the scope and lifetime of variables in Python with suitable examples. [7]

Q.4 (a) Write a Python program using for and while loops to generate a sequence of numbers from 1 to 10. Explain the working of the loops used. [7]

(b) Write a Python program to demonstrate commonly used string methods such as upper(), lower(), replace(), split() and join(). [7]

Q.5 (a) Write a Python program to perform basic operations on a list such as adding, removing, updating and accessing elements. Explain the operations used. [7]

(b) Explain list indexing and slicing in Python with suitable examples. Write a Python program to access and extract specific elements from a list using indexing and slicing. [7]

Q.6 (a) Consider the following dictionary: [7]
student = {"Name": "Shyam", "Age": 18, "Branch": "CSE", "Marks": 85}
Write Python statements to perform different operations such as accessing values, adding a new key-value pair, modifying a value, deleting an item and displaying all keys and values.

(b) Explain the built-in functions and methods available for tuples in Python with suitable examples. [7]

Q.7 (a) Write a Python program to create a text file, write data into it and read the contents of the file. Explain the file operations used. [7]

(b) Write a Python program to read a text file and count the number of lines, words and characters in the file. Explain the approach used. [7]

Q.8 (a) Explain Object-Oriented Design in Python. Discuss how OOP concepts are used in designing Python programs. [7]

(b) What is Inheritance? Explain single inheritance in Python with a suitable example. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]
(a) Operator Precedence and Associativity
(b) zip() Function in Python
(c) Binary Files in Python
(d) NumPy and Its Applications

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